Table 1: S. cerevisiae datasets (pre-build, 2026-10-15) built as LMDBs in the TorchCell dev tree, with a description of each. *Description* is the summary line of the dataset class's docstring; the other columns are as in the non-expanded table.

| Dataset | Host | Description | Genotypes | Env | Instances | Phenotype | Shape | Graph role | Signal (gzip, bits) |
|---|---|---|---|---|---|---|---|---|---|
| **Environmental / chemogenomic response** | | | | | | | | | |
| CellYieldDeChiara2022 | S. cerevisiae | Colony cell yield (cells) of 1,011 isolates in 43 environments. | 906 | 43 | 36,286 | environment_response (cell_yield) | scalar | global | $1.7 \times 10^{7}$ |
| CrisprMagicLian2019 | S. cerevisiae | Lian 2019 MAGIC per-guide CRISPRa/i/d furfural-enrichment env x geno dataset. | 266,304 | 3 | 266,304 | environment_response (log2_ratio) | scalar | global | $1.3 \times 10^{8}$ |
| CrispriChemgenSmith2016 | S. cerevisiae | Smith 2016 per-guide dCas9-Mxi1 CRISPRi chemical-genetic env x geno dataset. | 1,035 | 26 | 14,463 | environment_response (log2_ratio) | scalar | global | $7.5 \times 10^{6}$ |
| CrispriMormino2022 | S. cerevisiae | Mormino 2022 CRISPRi acetic-acid biosensor screen (12 isolated strains). | 12 | 1 | 12 | environment_response (categorical) | scalar | global | $1.4 \times 10^{4}$ |
| DoublingTimeDeChiara2022 | S. cerevisiae | Colony population doubling time (h) of 1,011 isolates in 43 environments. | 906 | 43 | 36,295 | environment_response (growth_rate) | scalar | global | $1.7 \times 10^{7}$ |
| EnvChemgenAuesukaree2009 | S. cerevisiae | Auesukaree 2009 six-stress env x geno -> categorical sensitivity dataset. | 333 | 6 | 525 | environment_response (categorical) | scalar | global | $3.1 \times 10^{5}$ |
| EnvChemgenCostanzo2021 | S. cerevisiae | Costanzo 2021 condition-SGA: env x (deletion \| ts-allele) -> differential fitness. | 4,422 | 14 | 61,640 | environment_response (differential_fitness) | scalar | global | $1.1 \times 10^{8}$ |
| EnvChemgenHoepfner2014 | S. cerevisiae | Hoepfner 2014 HIP-HOP atlas: env x (het \| hom kanMX4 deletion) -> sensitivity score. | 10,839 | 5,586 | 30,308,574 | environment_response (sensitivity_score) | scalar | global | $2.1 \times 10^{10}$ |
| EnvChemgenMota2024 | S. cerevisiae | Acetic/butyric/octanoic acid chemogenomic env x geno -> ordinal susceptibility. | 607 | 3 | 1,289 | environment_response (ordinal) | scalar | global | $7.4 \times 10^{5}$ |
| EnvChemgenParsons2006 | S. cerevisiae | Parsons 2006 chemical-genetic compendium: env x geno -> log2(control/treated). | 4,111 | 82 | 329,254 | environment_response (log2_ratio) | scalar | global | $2.2 \times 10^{8}$ |
| EnvChemgenPiotrowski2017 | S. cerevisiae | MOSAIC diagnostic-pool chemical genetics: env x geno -> CG interaction z-score. | 289 | 11,949 | 3,435,820 | environment_response (z_score) | scalar | global | $1.4 \times 10^{9}$ |
| EnvChemgenVanacloig2022 | S. cerevisiae | Anaerobic chemical-genomic env x geno -> log2(inhibitor/control) response screen. | 3,649 | 45 | 164,205 | environment_response (log2_ratio) | scalar | global | $9.3 \times 10^{7}$ |
| EnvChemgenWildenhain2015 | S. cerevisiae | Wildenhain 2015 chemical-genetic matrix: env x geno -> growth-inhibition z-score. | 258 | 5,178 | 449,033 | environment_response (z_score) | scalar | global | $2.6 \times 10^{8}$ |
| EnvChemgenYoshikawa2009 | S. cerevisiae | Yoshikawa 2009 growth rates: env x geno -> one culture's specific growth rate. | 4,729 | 4 | 37,832 | environment_response (growth_rate) | scalar | global | $1.0 \times 10^{7}$ |
| FattyAcidSmith2006 | S. cerevisiae | Smith 2006 fatty-acid clear-zone env x geno -> ordinal response dataset. | 4,770 | 3 | 14,310 | environment_response (ordinal) | scalar | global | $1.8 \times 10^{6}$ |
| HetHillenmeyer2008 | S. cerevisiae | FitDb HIP (heterozygous kanMX4 allele replacement) fitness-defect log2-ratio. | 17,659 | 512 | 3,074,132 | environment_response (log2_ratio) | scalar | global | $2.0 \times 10^{9}$ |
| HomHillenmeyer2008 | S. cerevisiae | FitDb HOP (homozygous barcoded kanMX4 deletion) fitness-defect z-score dataset. | 9,220 | 276 | 1,228,676 | environment_response (z_score) | scalar | global | $8.4 \times 10^{8}$ |
| IsolateGrowthPeter2018 | S. cerevisiae | Peter 2018: one record per (isolate, condition), 33,985 records. | 971 | 35 | 33,985 | environment_response (condition_growth_ratio) | scalar | global | $1.6 \times 10^{7}$ |
| **Expression (RNA-seq)** | | | | | | | | | |
| Boocock2025 | S. cerevisiae | Boocock 2025: 393 pooled segregants plus 79,123 de novo segregant cells. | 79,516 | 2 | 79,516 | umi_count (scrnaseq_umi_count) | vector (3797) | node | $8.9 \times 10^{9}$ |
| CaudalPanTranscriptome2024 | S. cerevisiae | Natural-isolate pan-transcriptome: each isolate a perturbation set off S288C. | 943 | 1 | 943 | expression_tpm (rnaseq_tpm) | vector (6017) | node | $1.7 \times 10^{9}$ |
| NadalRibellesPerturbSeq2025 | S. cerevisiae | Nadal-Ribelles 2025 pseudobulk Perturb-seq: one record per (genotype, condition). | 3,150 | 2 | 6,188 | expression_log2_ratio (pseudobulk_scrnaseq_log2fc) | vector (5818) | node | $2.5 \times 10^{9}$ |
| Nguessan2025 | S. cerevisiae | N'Guessan 2025: 2,951 BB-QTL batch-1 segregants, their cells summed. | 2,951 | 1 | 2,951 | umi_count (scrnaseq_umi_count) | vector (1670) | node | $4.5 \times 10^{8}$ |
| PooledPromoterExpressionVaishnav2022 | S. cerevisiae | Vaishnav 2022 GPRA sort-seq expression of 71,158 promoter sequences in SC-Ura. | 71,158 | 1 | 71,158 | expression_level | scalar | node | $1.3 \times 10^{8}$ |
| TfDeletionPseudobulkJackson2020 | S. cerevisiae | Jackson 2020: one record per (barcoded strain, condition), 784 records. | 72 | 11 | 784 | umi_count (scrnaseq_umi_count) | vector (5075) | node | $9.7 \times 10^{7}$ |
| **Expression (microarray)** | | | | | | | | | |
| DmMicroarraySameith2015 | S. cerevisiae | Double mutant GSTF expression profiling from Sameith et al. 2015. | 72 | 1 | 72 | expression_log2_ratio | vector (6169) | node | $1.7 \times 10^{8}$ |
| InductionTimecourseHackett2020 | S. cerevisiae | Hackett 2020 IDEA: one record per (time course, time point), 1,752 records. | 214 | 31 | 1,752 | log2_cleaned | vector (6175) | node | $4.7 \times 10^{9}$ |
| MicroarrayKemmeren2014 | S. cerevisiae | Deletion-mutant microarray expression profiling from Kemmeren et al. 2014. | 1,484 | 1 | 1,484 | expression_log2_ratio | vector (6169) | node | $3.2 \times 10^{9}$ |
| MicroarrayScoreHu2007 | S. cerevisiae | Hu 2007 TF-deletion transcriptomes as released: error-model X scores + p. | 263 | 1 | 263 | expression_score (error_model_x_score_weighted_duplicate_average) | vector (6371) | node | $2.5 \times 10^{8}$ |
| SmMicroarraySameith2015 | S. cerevisiae | Single mutant GSTF expression profiling from Sameith et al. 2015. | 82 | 1 | 82 | expression_log2_ratio | vector (6169) | node | $1.8 \times 10^{8}$ |
| **Fitness** | | | | | | | | | |
| Bloom2019 | S. cerevisiae | Bloom 2019: one record per (segregant, condition), 530,100 records. | 13,950 | 38 | 530,100 | environment_response (control_regression_residual) | scalar | global | $2.6 \times 10^{8}$ |
| CrispriSegregantHale2024 | S. cerevisiae | Hale 2024: one record per (double-barcode lineage, assay), 8,692,635 records. | 895,494 | 2 | 8,692,635 | environment_response (relative_fitness) | scalar | global | $6.7 \times 10^{9}$ |
| DmfCostanzo2016 | S. cerevisiae | Double-mutant fitness dataset from Costanzo et al. 2016. | 20,705,612 | 2 | 20,705,612 | fitness | scalar | global | $7.1 \times 10^{9}$ |
| DmfKuzmin2018 | S. cerevisiae | Double-mutant fitness experiments from Kuzmin 2018. | 410,571 | 1 | 410,571 | fitness | scalar | global | $1.5 \times 10^{8}$ |
| DmfKuzmin2020 | S. cerevisiae | Double-mutant fitness experiments from Kuzmin et al. 2020. | 608,805 | 1 | 632,998 | fitness | scalar | global | $2.4 \times 10^{8}$ |
| SmfBaryshnikova2010 | S. cerevisiae | Baryshnikova 2010 genome-scale single-mutant fitness dataset (6023 alleles). | 6,023 | 2 | 6,023 | fitness | scalar | global | $1.8 \times 10^{6}$ |
| SmfCostanzo2016 | S. cerevisiae | Single-mutant fitness dataset from Costanzo et al. 2016. | 10,277 | 2 | 20,484 | fitness | scalar | global | $5.4 \times 10^{6}$ |
| SmfKuzmin2018 | S. cerevisiae | Single-mutant fitness experiments from Kuzmin 2018. | 1,539 | 1 | 1,539 | fitness | scalar | global | $3.9 \times 10^{5}$ |
| SmfKuzmin2020 | S. cerevisiae | Single-mutant fitness experiments from Kuzmin et al. 2020. | 472 | 1 | 472 | fitness | scalar | global | $9.3 \times 10^{4}$ |
| SmfODuibhir2014 | S. cerevisiae | O'Duibhir 2014 single-deletion relative-growth-rate -> fitness dataset. | 1,312 | 1 | 1,312 | fitness | scalar | global | $2.0 \times 10^{6}$ |
| TmfKuzmin2018 | S. cerevisiae | Triple-mutant fitness experiments from Kuzmin 2018. | 91,111 | 1 | 91,111 | fitness | scalar | global | $3.9 \times 10^{7}$ |
| TmfKuzmin2020 | S. cerevisiae | Triple-mutant fitness experiments from Kuzmin et al. 2020. | 290,355 | 1 | 301,798 | fitness | scalar | global | $1.3 \times 10^{8}$ |
| **Gene interaction** | | | | | | | | | |
| DmiCostanzo2016 | S. cerevisiae | Double-mutant genetic interaction dataset from Costanzo et al. 2016. | 20,705,612 | 2 | 20,705,612 | gene_interaction | scalar | edge | $6.0 \times 10^{9}$ |
| DmiKuzmin2018 | S. cerevisiae | Double-mutant genetic-interaction experiments from Kuzmin 2018. | 410,399 | 1 | 410,399 | gene_interaction | scalar | edge | $1.3 \times 10^{8}$ |
| DmiKuzmin2020 | S. cerevisiae | Digenic interaction experiments from Kuzmin et al. 2020. | 608,604 | 1 | 632,797 | gene_interaction | scalar | edge | $2.1 \times 10^{8}$ |
| TmiKuzmin2018 | S. cerevisiae | Triple-mutant genetic-interaction experiments from Kuzmin 2018. | 91,111 | 1 | 91,111 | gene_interaction | scalar | hyperedge | $3.4 \times 10^{7}$ |
| TmiKuzmin2020 | S. cerevisiae | Trigenic interaction experiments from Kuzmin et al. 2020. | 290,355 | 1 | 301,798 | gene_interaction | scalar | hyperedge | $1.1 \times 10^{8}$ |
| **Genome architecture** | | | | | | | | | |
| GenomeArchitecturePuddu2019 | S. cerevisiae | Puddu 2019: one record per sequenced knockout colony, 8,835 records. | 4,550 | 1 | 8,835 | genome_architecture | vector (28) | global | $1.2 \times 10^{7}$ |
| **Life cycle** | | | | | | | | | |
| LifeCycleDeChiara2022 | S. cerevisiae | Sporulation and stationary-phase survival time courses of 1,011 isolates. | 905 | 5 | 4,039 | life_cycle | vector (4) | global | $1.3 \times 10^{6}$ |
| **Metabolite** | | | | | | | | | |
| AminoAcidCooper2010 | S. cerevisiae | Cooper 2010 CE-LIF amino-acid pools of the deletion collection (ratio to plate). | 4,334 | 1 | 4,382 | metabolite_level (ce_lif_peak_area_ratio_to_plate_mean) | vector (16) | bipartite node | $5.2 \times 10^{6}$ |
| AminoAcidMulleder2016 | S. cerevisiae | Genome-wide intracellular amino-acid metabolome of the yeast deletion collection. | 4,678 | 1 | 4,678 | metabolite_level (intracellular_concentration_mM) | vector (19) | bipartite node | $1.1 \times 10^{7}$ |
| BetaxanthinCachera2023 | S. cerevisiae | Genome-wide betaxanthin CRI-SPA product-proxy screen of the YKO collection. | 4,759 | 1 | 4,759 | metabolite_level (cri_spa_corrected_hsv_yellowness_24h) | scalar | bipartite node | $4.5 \times 10^{6}$ |
| FattyAcidXue2025 | S. cerevisiae | Xue 2025 in-house FFA combinatorial-deletion titer dataset (5 FFAs/strain). | 176 | 1 | 176 | metabolite_level (titer_mg_per_l) | vector (5) | bipartite node | $1.9 \times 10^{5}$ |
| IsobutanolScreenLopez2024 | S. cerevisiae | Montaño López 2024 FIRST genome-wide isobutanol-biosensor YKO screen (Table S2, n=1). | 4,607 | 1 | 4,805 | metabolite_level (biosensor_gfp_fluorescence_fold_change) | scalar | bipartite node | $9.0 \times 10^{5}$ |
| IsobutanolValidatedLopez2024 | S. cerevisiae | Montaño López 2024 VALIDATED isobutanol-biosensor re-screen (Table S3, triplicate). | 226 | 1 | 227 | metabolite_level (biosensor_gfp_fluorescence_fold_change) | scalar | bipartite node | $7.4 \times 10^{4}$ |
| MetaboliteDaSilveira2014 | S. cerevisiae | MS lipidome of the yeast kinase/phosphatase deletion collection (127 mutants). | 127 | 1 | 127 | metabolite_level (lipidomics_ms_relative_abundance_au) | vector (135) | bipartite node | $1.1 \times 10^{6}$ |
| MetaboliteZelezniak2018 | S. cerevisiae | LC-SRM targeted metabolome of the kinase-knockout collection, one record per (strain, protocol); each record and its WT reference share one protocol. | 95 | 1 | 129 | metabolite_level (lc_srm_signal_batch_corrected_uncalibrated_dataset1) | vector (14) | bipartite node | $3.5 \times 10^{5}$ |
| OrganicAcidYoshida2012 | S. cerevisiae | HPLC organic-acid titers of 17 yeast deletion mutants (Yoshida 2012, Table 3). | 17 | 1 | 17 | metabolite_level (hplc_organic_acid_titer_mM) | vector (6) | bipartite node | $3.3 \times 10^{4}$ |
| **Morphology** | | | | | | | | | |
| CarotenoidOzaydin2013 | S. cerevisiae | Beta-carotene colony-color visual screen of the yeast deletion collection. | 4,475 | 1 | 4,607 | visual_score | scalar | global | $7.7 \times 10^{6}$ |
| ScmdOhnuki2018 | S. cerevisiae | CalMorph morphology of essential-gene heterozygous diploid deletions (Ohnuki 2018). | 1,112 | 1 | 1,112 | calmorph | vector (281) | global | $4.9 \times 10^{7}$ |
| ScmdOhnuki2022 | S. cerevisiae | CalMorph morphology for 1979 quadruple (target + 3Delta) deletion strains. | 1,982 | 1 | 1,982 | calmorph | vector (281) | global | $8.8 \times 10^{7}$ |
| ScmdOhya2005 | S. cerevisiae | CalMorph morphology for haploid non-essential single-gene deletions (Ohya 2005). | 4,718 | 1 | 4,718 | calmorph | vector (281) | global | $1.5 \times 10^{8}$ |
| **Protein abundance** | | | | | | | | | |
| ProteomeMessner2021 | S. cerevisiae | Scanning SWATH proteomes of wild-type yeast under 18 compounds vs DMSO (70 runs). | 1 | 18 | 70 | protein_abundance (swath_ms_diann_maxlfq_proteotypic_quantity) | vector (1920) | node | $1.2 \times 10^{7}$ |
| ProteomeMessner2023 | S. cerevisiae | Microflow-SWATH-MS proteome of the yeast gene-KO collection (4,699 strains). | 4,549 | 1 | 4,699 | protein_abundance (swath_ms_maxlfq_batch_corrected_quantity) | vector (1830) | node | $1.2 \times 10^{9}$ |
| ProteomeMuenzner2024 | S. cerevisiae | 796 SWATH-MS proteomes (SI Table 3): 794 isolates with a Peter genome and FY4. | 795 | 1 | 795 | protein_abundance (swath_ms_maxlfq_precursor_batch_corrected_quantity) | vector (1542) | node | $9.6 \times 10^{8}$ |
| ProteomeZelezniak2018 | S. cerevisiae | SWATH-MS proteome of the yeast kinase-knockout collection (97 strains). | 97 | 1 | 97 | protein_abundance (swath_ms_label_free_log_signal_sva) | vector (726) | node | $1.7 \times 10^{7}$ |
| **Protein synthesis / turnover** | | | | | | | | | |
| ProteinTurnoverMuenzner2024 | S. cerevisiae | Dynamic-SILAC turnover rates for 55 natural isolates (SI Table 17). | 55 | 1 | 55 | degradation_rate (dynamic_silac_lys8_kdp_per_hour_no_dilution_correction) | vector (1015) | node | $6.9 \times 10^{7}$ |
| **Viability** | | | | | | | | | |
| GeneEssentialitySgd | S. cerevisiae | Gene essentiality experiments built from SGD inviable null phenotypes. | 1,180 | 5 | 1,329 | is_essential | scalar | node | $6.8 \times 10^{5}$ |
| SynthLethalityYeastSynthLethDb | S. cerevisiae | Yeast synthetic lethality gene-pair experiments from SynLethDB. | 14,000 | 1 | 14,000 | is_synthetic_lethal | scalar | edge | $8.0 \times 10^{6}$ |
| SynthRescueYeastSynthLethDb | S. cerevisiae | Yeast synthetic rescue gene-pair experiments from SynLethDB. | 6,948 | 1 | 6,948 | is_synthetic_rescue | scalar | edge | $4.0 \times 10^{6}$ |
| **Total (69 datasets, 69 probed)** | | | | | **93,266,786** | | | | $7.2 \times 10^{10}$ |